

Semester	T.E. Semester V– BIOM
Subject	Biomedical Digital Signal Processing
Subject Professor In-charge	Prof. Geetha Narayanan
Student Name	Rehan Raza Shaikh
Roll Number	24105A0037
Grade and Subject Teacher's Signature	

Week 4 Submission	
Project Title	Phonocardiogram (PCG)
Weekly Objective	Discrete Fourier Transform , Signal Frequency , Add Noise , Powerline Noise,Original vs Powerline Noise
Key concepts learned	DFT , Powerline Noise 50 HZ ,
Work Completed	DFT , Powerline Noise , Comparission
Code	<pre> import numpy as np import matplotlib.pyplot as plt # Select first 1000 samples samples = 1000 x = pcg[:samples] print("Number of samples used:", len(x)) </pre>

```
# Compute DFT using FFT
```

```
X = np.fft.fft(x)
```

```
# Number of samples
```

```
N = len(x)
```

```
# Frequency axis
```

```
frequency = np.fft.fftfreq(N, d=1/fs)
```

```
# Keep only positive frequencies
```

```
positive = frequency >= 0
```

```
# =====
```

```
# Magnitude Spectrum
```

```
# =====
```

```
plt.figure(figsize=(12,5))
```

```
plt.plot(frequency[positive], np.abs(X)[positive])
```

```
plt.title("DFT Magnitude Spectrum of First 1000 PCG Samples")
```

```
plt.xlabel("Frequency (Hz)")
```

```
plt.ylabel("Magnitude")
```

```
plt.grid(True)
```

```
plt.show()
```

```
# =====
```

```
# Phase Spectrum
```

```
# =====
```

```
plt.figure(figsize=(12,5))

plt.plot(frequency[positive], np.angle(X)[positive])

plt.title("DFT Phase Spectrum of First 1000 PCG Samples")

plt.xlabel("Frequency (Hz)")

plt.ylabel("Phase (Radians)")

plt.grid(True)

plt.show()

magnitude = np.abs(X)[positive]

freq = frequency[positive]

peak = np.argmax(magnitude)

print(f"Dominant Frequency: {freq[peak]:.2f} Hz")

print(f"Magnitude: {magnitude[peak]:.4f}")

# =====

# First 10 DFT Coefficients

# =====

print("\nFirst 10 DFT Coefficients:")

for i in range(10):

    print(f"X[{i}] = {X[i]}")

import numpy as np

import matplotlib.pyplot as plt

# =====
```

```
# First 1000 Samples

# =====

samples = 1000

x = pcg[:samples]

# =====

# Compute DFT (FFT)

# =====

X = np.fft.fft(x)

N = len(x)

# Frequency axis

frequency = np.fft.fftfreq(N, d=1/fs)

# Keep only positive frequencies

positive = frequency >= 0

freq = frequency[positive]

magnitude = np.abs(X)[positive]

# =====

# Remove DC Component (0 Hz)

# =====

magnitude[0] = 0

# =====
```

```
# Dominant Frequency

# =====

peak_index = np.argmax(magnitude)

dominant_frequency = freq[peak_index]
dominant_magnitude = magnitude[peak_index]

print(f"Dominant Frequency = {dominant_frequency:.2f} Hz")
print(f"Peak Magnitude = {dominant_magnitude:.2f}")

# =====

# Plot Frequency Spectrum
# =====

plt.figure(figsize=(12,5))
plt.plot(freq, magnitude)
plt.scatter(dominant_frequency, dominant_magnitude,
            color='red', label='Dominant Frequency')
plt.title("Frequency Spectrum (First 1000 PCG Samples)")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.legend()
plt.grid(True)
plt.show()

import numpy as np
import matplotlib.pyplot as plt
```

```
# =====  
  
# Select first 1000 samples  
  
# =====  
  
samples = 1000  
x = pcg[:samples]  
  
  
# =====  
  
# Generate 50 Hz Power Line Noise  
  
# =====  
  
t = np.arange(samples) / fs    # Time vector  
  
  
noise_frequency = 50          # 50 Hz (India)  
noise_amplitude = 0.05       # Adjust amplitude if needed  
  
  
powerline_noise = noise_amplitude * np.sin(2 * np.pi * noise_frequency *  
t)  
  
  
# =====  
  
# Add Noise to PCG Signal  
  
# =====  
  
noisy_pcg = x + powerline_noise  
  
  
# =====  
  
# Plot Original and Noisy Signals  
  
# =====  
  
plt.figure(figsize=(14,6))
```

```
plt.subplot(2,1,1)

plt.plot(t, x)

plt.title("Original PCG Signal (1000 Samples)")

plt.xlabel("Time (s)")

plt.ylabel("Amplitude")

plt.grid(True)


plt.subplot(2,1,2)

plt.plot(t, noisy_pcg, color='red')

plt.title("PCG Signal with 50 Hz Power Line Noise")

plt.xlabel("Time (s)")

plt.ylabel("Amplitude")

plt.grid(True)


plt.tight_layout()

plt.show()


# =====

# Plot Only the Power Line Noise

# =====

plt.figure(figsize=(12,4))

plt.plot(t, powerline_noise, color='green')

plt.title("50 Hz Power Line Noise")

plt.xlabel("Time (s)")

plt.ylabel("Amplitude")

plt.grid(True)
```

```
plt.show()

# =====

# DFT of Noisy PCG Signal (1000 Samples)

# =====

import numpy as np

import matplotlib.pyplot as plt

# noisy_pcg should already contain:

# noisy_pcg = x + powerline_noise

# Compute DFT

X = np.fft.fft(noisy_pcg)

# Number of samples

N = len(noisy_pcg)

# Frequency axis

frequency = np.fft.fftfreq(N, d=1/fs)

# Keep only positive frequencies

positive = frequency >= 0

freq = frequency[positive]

dft = np.abs(X)[positive]
```



```
# Remove DC component for better visualization (optional)

dft[0] = 0

# =====

# Plot DFT

# =====

plt.figure(figsize=(12,5))

plt.plot(freq, dft, color='blue', linewidth=2)

plt.title("Discrete Fourier Transform (DFT) of Noisy PCG Signal")

plt.xlabel("Frequency (Hz)")

plt.ylabel("Magnitude")

plt.grid(True)

plt.show()

# =====

# Dominant Frequency

# =====

peak = np.argmax(dft)

print("Dominant Frequency :", freq[peak], "Hz")

print("Maximum Magnitude :", dft[peak])

# =====

# Print First 20 DFT Coefficients
```

```
# =====

print("\nFirst 20 DFT Coefficients (Magnitude):")

for i in range(20):

    print(f"X[{i}] = {abs(X[i]):.4f}")

    #
=====

# Compare Original PCG and Noisy PCG (1000 Samples)

# =====

import matplotlib.pyplot as plt
import numpy as np

# Time axis
t = np.arange(len(x)) / fs

plt.figure(figsize=(15,6))

# Original PCG
plt.plot(t, x,

        label='Original PCG Signal',

        linewidth=2)

# Noisy PCG
plt.plot(t, noisy_pcg,

        label='PCG + 50 Hz Power Line Noise',
```

```
linewidth=1.5,  
  
alpha=0.8)  
  
plt.title("Comparison of Original and Noisy PCG Signal")  
plt.xlabel("Time (seconds)")  
plt.ylabel("Amplitude")  
plt.legend()  
plt.grid(True)  
  
plt.show()  
  
# =====  
  
# Original vs Noisy PCG (Separate Plots)  
  
# =====  
  
plt.figure(figsize=(15,8))  
  
# Original Signal  
plt.subplot(2,1,1)  
plt.plot(t, x, color='blue')  
plt.title("Original PCG Signal")  
plt.xlabel("Time (seconds)")  
plt.ylabel("Amplitude")  
plt.grid(True)  
  
# Noisy Signal  
plt.subplot(2,1,2)
```

```
plt.plot(t, noisy_pcg, color='red')

plt.title("PCG Signal After Adding 50 Hz Power Line Noise")

plt.xlabel("Time (seconds)")

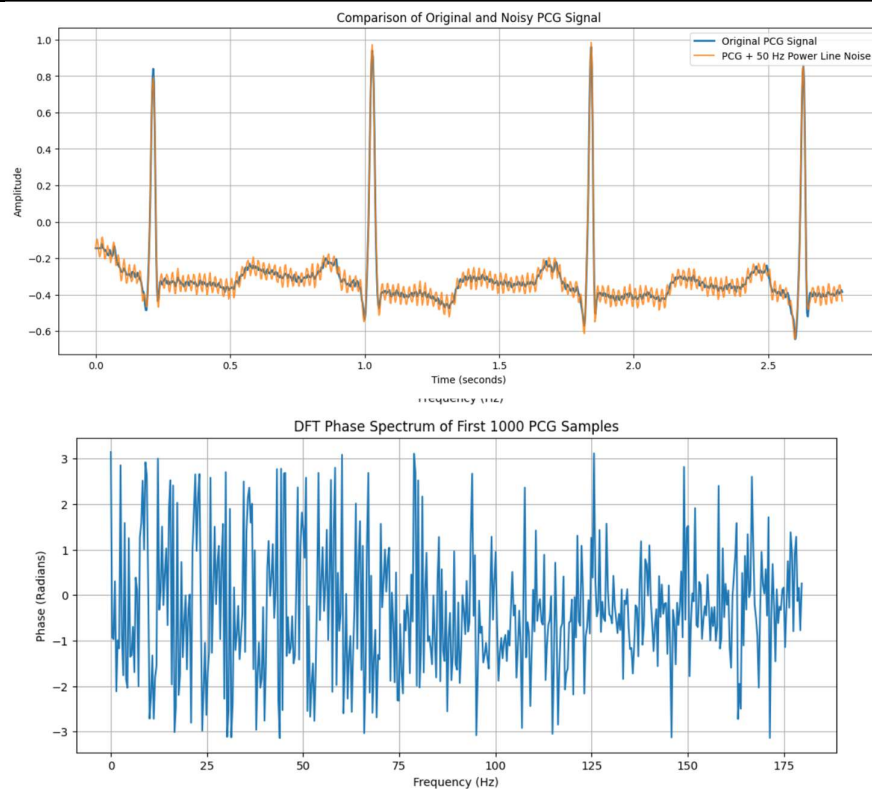
plt.ylabel("Amplitude")

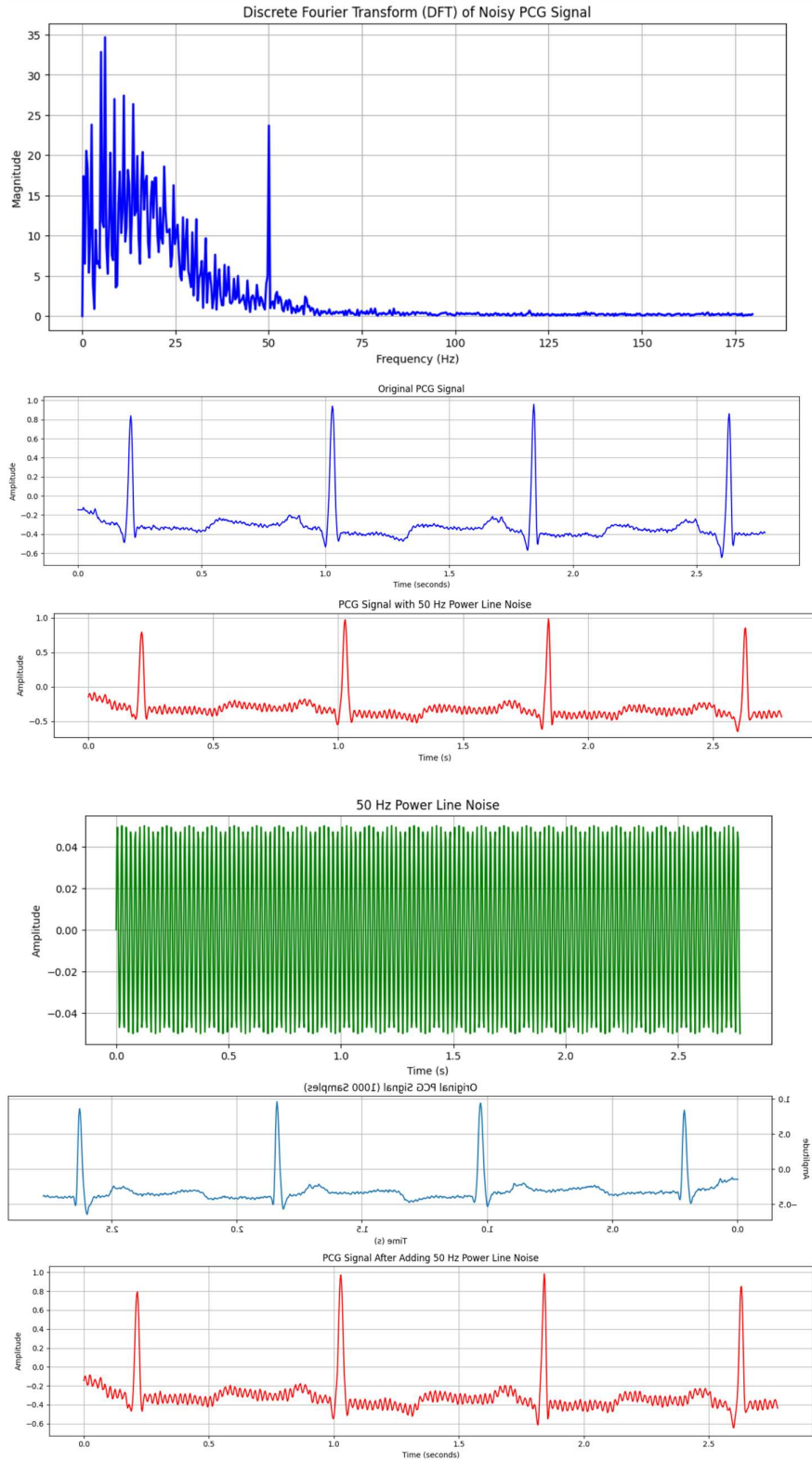
plt.grid(True)

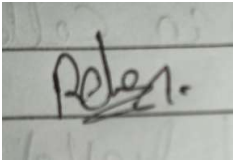
plt.tight_layout()

plt.show()
```

Evidence of
Progress/Result
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	DFT Magnitude Spectrum of First 1000 PCG Samples
Challenges & Solutions	Firstly we face wrong data in graph then , then we use 1000 sample data and do all works
Next Week Plan	Week 5
Declaraion	I hereby declare that for the assignment / academic activity submitted by me, I confirm ONE of the following statements by ticking (☐) the appropriate box: ☐ (1) I have prepared the assignment / academic activity entirely by myself. ☐ (2) I have completed the assignment / academic activity with academic guidance or discussion from seniors / friends / peers, while ensuring the work is my own. ☐ (3) I have used AI-based tools only as an aid and have appropriately acknowledged or cited their use as per the ethical guidelines of the Institute. ☐ (4) I have used AI-based tools and directly copy-pasted content for the assignment / academic activity. ☐ (5) I have copied the assignment / academic activity from peers / friends. I understand that selecting options (4) or (5) indicates unethical academic practice and may attract academic penalties, including rejection of the assignment or disciplinary action, as per the rules of Vidyalankar Institute of Technology, Mumbai. I submit this declaration truthfully and accept full responsibility for the same. Student Name: Rehan Raza Shaikh Roll No.: 24105A0037  Signature: Date:05/08/2026

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